CO2 Footprint Tracker(Requirements)

1. **Functional Requirements:**

- **User Stories:**

    1. Setting Up a Profile:
        - "As a user, I want to create a lifestyle profile that allows for a personalized calculation of my carbon footprint."
    2. Logging Activities:
        - "As a user, I want to record my daily activities to understand how they influence my CO_2 emissions."
    3. Analyzing Emissions:
        - "As a user, I want to see a categorized overview of my CO_2 emissions to identify which aspects of my lifestyle contribute the most."
    4. Receiving Recommendations:
        - "As a user, I want to get personalized suggestions for reducing my carbon footprint based on the data I have logged."

- **Use Cases:**
    - Created use case diagrams that depict key interactions between the user and the application.
    - Important use cases consist of:
        - User profile configuration
        - Daily activity logging
        - CO_2 emissions analysis
        - Receiving suggestions for optimization

2. **Non-Functional Requirements**

- Performance:
    - The application should respond promptly and handle user inputs efficiently to provide a smooth experience.
- Data Reliability:
    - The calculations for CO_2 emissions must be dependable, using established emission factors for various activities.
- User Experience:
    - An easy-to-navigate and intuitive user interface is crucial for encouraging consistent tracking and logging of activities by users.

3. **Constraints**:

- Database Availability:

Due to the unavailability of MySQL in the lab environment, alternative data storage solutions will be utilized.

- Data Storage Options:

We will implement **SQLite** as a lightweight relational database or utilize **JSON files** for managing user and activity data.